

TEAM-02 · TCP IP Communication- INDIVIDUAL CONTRIBUTION EVIDENCE REPORT

Student Information

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Project: How Information Travels Across a Network Using the TCP/IP Model

Contribution: Technical research, presentation preparation and delivery

Introduction

For this project, we were required to prepare and deliver a technical presentation explaining how information travels across a network using the TCP/IP model.

Our original team was expected to consist of five members. However, only two of us participated in the preparation and delivery of the presentation. We therefore worked together to complete the project and divide the presentation responsibilities between us.

I was responsible for preparing and presenting the second part of our presentation. This report documents the sections I worked on and presented as evidence of my individual contribution.

My Assigned Sections

I was responsible for the following sections of our presentation:

- Encapsulation and decapsulation
- TCP/IP and OSI model mapping
- TCP and UDP comparison
- Packet journey across the network
- Network communication troubleshooting
- Conclusion

I worked on these sections with the aim of making the technical concepts clear and easy to understand during our presentation.

3. Encapsulation and Decapsulation

I explained the process of encapsulation and decapsulation and how they allow data to move through the network.

We explained that during encapsulation, application data moves down through the TCP/IP layers. As the data passes through the layers, additional information is added to help with communication and delivery.

At the receiving device, the reverse process takes place. The information added during transmission is processed as the data moves back up through the layers. We referred to this process as decapsulation.

4. TCP/IP and OSI Model Mapping

I presented how the four TCP/IP layers relate to the seven layers of the OSI model.

In our presentation, we showed the following mapping:

TCP/IP Layer	Corresponding OSI Layers
Application	Application, Presentation and Session
Transport	Transport
Internet	Network
Network Access	Data Link and Physical

I explained that the TCP/IP model combines some functions that are represented as separate layers in the OSI model.

TCP and UDP Comparison

I presented our comparison of TCP and UDP and explained why different applications may use different transport protocols.

We explained that TCP is connection-oriented and provides reliable communication using mechanisms such as acknowledgements and retransmission.

We also explained that UDP is connectionless and has lower overhead because it does not provide the same built-in reliability mechanisms as TCP.

In our presentation, we used the following examples:

TCP applications:

- HTTPS/HTTP
- Email services

UDP applications:

- DNS
- Real-time voice and video applications

Packet Journey Diagram

I presented the packet journey diagram that we created to show how information travels from a sending device to a destination.

Our diagram showed the movement of data through network devices and the direction of communication between the client and destination server.

I also explained how different types of addressing are involved during communication, including IP addresses, MAC addresses and port numbers.

Network Troubleshooting

I presented the troubleshooting section of our project. We used a realistic communication failure to demonstrate how a network administrator could investigate and resolve a problem.

We explained that troubleshooting should be carried out systematically rather than making changes without first identifying the possible cause.

Our troubleshooting process included:

1. Checking the physical or wireless connection.
2. Checking the device's IP configuration.
3. Testing connectivity to the default gateway.
4. Testing connectivity to an external address.
5. Checking DNS resolution.
6. Checking relevant services and ports.
7. Identifying and correcting the configuration problem.
8. Testing the connection again after making the correction.

This helped us demonstrate how the TCP/IP model can be applied when diagnosing a real network communication problem.

Conclusion

I presented the conclusion of our presentation by summarising what we had covered about TCP/IP network communication.

We explained that communication across a network involves different layers, protocols, addressing methods and network devices working together.

Through our presentation, we showed how data moves from the sender to the destination through encapsulation and how it is processed through decapsulation at the receiving end.

My Contribution to the Group

I contributed to the research, preparation and delivery of the sections assigned to me. I worked with my teammate to organise the presentation and ensure that our technical explanations were clear and connected.

I also contributed to the preparation of the packet journey explanation and troubleshooting section. During the final presentation, I delivered my allocated sections and participated in the conclusion.

Although our original group was expected to have five members, only two of us participated in completing and presenting the project. We therefore shared the preparation and presentation responsibilities between us.

summary

Through this project, I helped prepare and deliver our technical presentation on TCP/IP network communication.

I presented encapsulation and decapsulation, TCP/IP and OSI model mapping, TCP and UDP comparison, the packet journey, troubleshooting, and the conclusion.

Working on this project helped me better understand how network layers, protocols, addressing, and network devices work together to enable communication between devices. It also gave us practical experience in explaining technical networking concepts as a team.